

Column1	Column2	Column3	Column4	Column5	Column6
MODEL comparison matrix					
Classification	Accuracy	Precision	Recall	F1Score	ROC/AUC
Logistic Regression	0.735795	0.501099	0.814286	0.620408	0.844567
Decision Tree Classifier	0.722538	0.477193	0.485714	0.481416	0.646189
Random Forest Classifier	0.774621	0.575000	0.575000	0.575000	0.826420
Gradient Boosting Classifier (e.g., XGBoost or LightGBM)	0.750000	0.517699	0.835714	0.639344	0.845501
K-Nearest Neighbors Classifier	0.682765	0.44086	0.732143	0.550336	0.767905
Support Vector Classifier (SVC)	0.734848	0.500000	0.825000	0.622642	0.831878
Voting Ensemble (RandomForest/GradientBoosting/LogisticRegression) - Validation Data	0.756629	0.527981	0.775000	0.628075	0.846827
Average Ensemble (RandomForest/GradientBoosting/LogisticRegression) - Validation Data	0.756629	0.527981	0.775000	0.628075	0.846827
Bayesian Ensemble Model	0.698864	0.462891	0.598485	0.846429	0.803652
Voting Ensemble (RandomForest/GradientBoosting/LogisticRegression) - Test Data	0.774834	0.558583	0.729537	0.632716	0.839922
Average Ensemble (RandomForest/GradientBoosting/LogisticRegression) - Test Data	0.774834	0.558583	0.729537	0.632716	0.839922

Which model gave you the best result and which was the worst?

Both the voting classifier and the average ensemble gave me equal results.

Indicating that the training data was similar, which in fact was.

Therefore, neither would be better or worse than the other.

However, I chose to compare my voting ensemble to the Bayesian ensemble model and the voting classifier ensemble model performed better than the Bayesian model.